

FUAD AHMED

MOTIVATION LETTER — MASTER'S DEGREE IN COMPUTER ENGINEERING (INGEGNERIA INFORMATICA)

Target Institution: Università degli Studi di Padova (DEI), Padua, Italy • **Intake:** Academic Year 2027/2028 (Fall 2027)
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Legal Name: TUBA, FUAD AHMED

To the Academic Evaluation Committee for Computer Engineering at the University of Padua,

I am writing to formally submit my application for admission to the **Master's Degree in Computer Engineering (Ingegneria Informatica)** at the **Università degli Studi di Padova** for the 2027/2028 academic year. As a software systems engineer with four years of hands-on professional experience building resilient backend architectures, high-concurrency microservices, and distributed data systems, my goal is to deepen my theoretical foundations and rigorous engineering competence under the mentorship of the prestigious Department of Information Engineering (DEI). The University of Padua's exceptional historical prestige, coupled with its forward-looking research into distributed systems, edge computing, and networked architectures, represents the ideal environment to conduct my Master's studies.

1. ACADEMIC TRAJECTORY & SYSTEMS FOUNDATIONS

I graduated from St. Mary's University in Addis Ababa with a Bachelor of Science in Computer Science, earning a Cumulative GPA of 3.20/4.00 within a rigorous four-year program conducted entirely in English. My undergraduate coursework provided me with solid mathematical competence across Discrete Mathematics, Linear Algebra, Probability & Statistics, and Calculus, alongside core systems disciplines including Operating Systems, Computer Architecture, Distributed Systems, Database Management Systems, and Advanced Algorithms.

For my senior capstone project, I led the architecture and empirical evaluation of a fault-tolerant, high-concurrency distributed data synchronization protocol. Operating across networked Linux nodes, my protocol utilized delta-encoded state serialization and lock-free memory ring buffers in Go and Python, ensuring state consistency across disparate nodes while drastically diminishing network payload overhead and CPU thread contention. This capstone solidified my ambition to specialize in large-scale distributed computing systems.

2. PROFESSIONAL ENGINEERING EXPERIENCE & MOTIVATION

Over the past four years, I have worked as a Lead Backend and Distributed Systems Software Engineer, developing enterprise-grade microservices and data pipelines using Python (FastAPI/AsyncIO), Go, and Docker. In production environments, I architected distributed Redis caching topologies that reduced database read latency by 48% and implemented asynchronous message-driven architectures using Kafka and RabbitMQ to guarantee system fault-tolerance during sudden load spikes.

While industry practice has honed my practical capabilities in Linux kernel tuning, container orchestration, and concurrent network programming, it has also revealed the necessity of deep theoretical mastery. Modern distributed computing confronts fundamental limits in consensus scalability, tail-latency bounds, and memory consistency models. The University of Padua's rigorous balance between formal computer science theory and rigorous engineering implementation provides the ideal crucible to master these challenges.

3. WHY THE UNIVERSITY OF PADUA & DEPARTMENT OF INFORMATION ENGINEERING (DEI)

Founded in 1222, the University of Padua is one of the world's most venerated centers of scientific inquiry, home to Galileo Galilei and a tradition of fearless intellectual rigor. Within modern engineering, the Department of Information Engineering (DEI) stands as an international leader in telecommunications, networked computing, and autonomous software systems.

I am particularly eager to engage with the advanced curriculum in Computer Engineering, specifically focusing on *Distributed Systems, Cloud Computing & Edge Networking, Advanced Computer Architectures, and Software Architecture*. Furthermore, I am deeply inspired by the DEI's active research in the cloud-to-edge computing continuum, decentralized consensus, and low-latency network telemetry. Collaborating with faculty who bridge foundational systems research with European industrial innovation is the exact environment where I can thrive.

4. CAREER VISION & LONG-TERM OBJECTIVES

Upon earning my Master's degree in Computer Engineering from the University of Padua, I plan to contribute as a Distributed Systems Architect within European technological enterprises or research laboratories, designing resilient, low-latency cloud and edge infrastructures. In the long term, I intend to leverage this elite European training to contribute to the modernization of distributed digital infrastructure in emerging economies across Africa.

I bring to the University of Padua proven engineering perseverance, strong academic discipline, and a deep respect for the university's scholarly tradition. I am fully prepared to meet the demanding standards of the MSc in Computer Engineering and contribute actively to the DEI student community.

Sincerely and respectfully,

Fuad Ahmed (TUBA, FUAD AHMED)

BSc in Computer Science, St. Mary's University

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